

Full name: Ha Hoang PhucMSSV: 222683Class: DH22TIN07LESSON 1:
Library management system-Identify candidate classes (Books, Readers,
Vouchers, Employees, Publishers)+ Book: main entity, managing book
information.

- + DocGia: person who borrows books.
- + PhieuMuon: records borrowing/returning.
- + NhanVien: system manager (librarian, warehouse keeper...).
- + NhaXuatBan: book source information.
 - Properties and behavior (methods) of each class + Book
 - oAttributes: maSach (PK, string), nameSach (string), tacGia (string),namXuatBan (int), manNXB (FK), soLuongTon (int), trangThai (string: "Yesavailable"/"On loan").
 - oBehavior: addSach(), capNhatSach(), deleteSach(), saveSach(),capNhatTonKho ().
- + DocGia
 - o Attributes: maDocGia (PK, string), hoTen (string), daySinh (date), diaChi (string), soDienThoai (string), dayDangKy (date), gioiTinh (boolean).
 - o Behavior: dangKyThe(), capNhatThongTin(), viewLichSuMuon(). +
- PhieuMuon
 - oAttributes: maPhieu (PK, string), maDocGia (FK), maSach (FK),FK, you're (date), you're (date), you're ThucTe(date), TinhTrang (string: "On loan"/"Paid"/"Overdue").
 - o Behavior: createPhieuMuon(), searchSach(), saveTraQuaHan(), crystalPhieuPhat(). +
- NhanVien
 - oAttributes: maNhanVien (PK, string), hoTen (string), chucVu (string),ngaySinh (date), soDienThoai (string).
 - o Behavior: dangNhap (), quanLyDocGia (), quanLyMuonTra (), quanLySach ().().
- + NhaXuatBan
 - o Attributes: maNXB (PK, string), nameNXB (string), diaChi (string), email (string).
 - oBehavior: addNXB(), capNhatNXB().
- RelationshipsBooks and Publishers: 1 book – 1 publisher

- +Readers and Vouchers: 1 reader – many borrowing slips
- +Books and Borrowing Tickets: 1 book – many borrowing coupons
- +Employees and Loan Vouchers: 1 employee – many vouchers

LESSON 2: Online sales management system

- Identify main use cases

Main actors:

- + Customer (Customers / Potential Customers).
- + NguoiQuanLy / Admin (Administrator / Sales staff).

Main use cases (customer registration, ordering, payment, product management):

KhachHang:

- +DangKyTaiKhan (Customer registration).
- +DangNhap (Sign in).
- +SeeSanPham (Search / View product list).
- +ThemVaoGioHang (Add to cart).
- +DatHang (Order).
- +ThanhToan (Online payment / COD).
- +View DonHang (View order history).

Admin / NguoiQuanLy:

- +QuanLySanPham (Add/Edit/Delete products).
- +QuanLyDonHang (View/Confirm/Cancel order).
- +QuanLyKhachHang (Customer information management).
- +QuanLyThanhToan (Payment processing, refund if necessary).

Extend / Include:

- +ThanhToan includes XacNhanDonHang.
- +DatHang extends ThanhToan (if paying online).

- From use case, find the necessary classes

- +Customers (Customer): from register, view orders.
- +SanPham (Product): product management.
- +GioHang (Cart): temporarily store products before ordering.
- +DonHang (Order): from order, payment.
- +ChiTietDonHang (OrderDetail): product details in the order.
- +ThanhToan (Payment): payment processing (may have subclasses such as OnlinePayment, COD).
- +Admin or NguoiQuanLy (inherits from User if there is a user system).

- Analyze relationships between classes (specify inheritance if any)KhachHang and DonHang: 1 customer – many ordersDonHang and ChiTietDonHang: 1 order – many order details (Composition)ChiTietDonHang and SanPham: many single details – 1 productDonHang and ThanhToan: 1 order – 1 paymentUser (abstract class) ←inherit ←KhachHang and Admin (KhachHang andAdmins are all the same type of User, sharing the same account, password, and full name).

Lesson 3: Convert class diagrams into data tables

1.MonHoc Table

No	School name	Data Types	Bind			
			PK	FK	Not NULL	Unique
1	maMon	Nvarchar(10)	x		x	x
2	tenMon	Nvarchar(50)			x	
3	soTinChiLT	Int			x	
4	soTinChiTH	Int			x	

2.Science Table

No	School name	Data Types	Bind			
			PK	FK	Not NULL	Unique
1	maHoHoc	Nvarchar(10)	x			x
2	tenKhoaHoc	Nvarchar(50)			x	
3	ngayTao	Date			x	